



NSA Christmas

Special announcement by the NSA during Christmas time. See what it is, on: <http://digit.in/NSAXMas>



Comic-Con Mumbai

Mumbai Film and Comic Convention 2014 gets a visit from the Digit Team <http://digit.in/1fAFnkr>

Digit 101

Digit 101 - Smartphone sensors

All the voodoo magic that your phone is capable of carrying out is by using some very smart sensors and here's how they work.

Ever wondered how your smartphone switches off the screen when you pull it close to your ears to answer a call? Or how the screen rotates on its own to re-orient itself in landscape/portrait mode? It all comes down to a few discrete sensors hidden away in your phone. Chances are that you'll never see them if you own a black coloured phone but with a slew of models now available in white fronts these "dots" have had the users intrigued. New features like cancelling a call by waving your hand over the phone is facilitated by these little sensors. Read on to know what all sensors your phone might have and how they come into play enabling certain features on your phone.

Accelerometer

Function: The accelerometer reads the acceleration of the device along three axes - X, Y and Z.

Feature: The accelerometer senses any tilt or motion and conveys it to the operating system. So when you tilt your phone or flick your phone it is this sensor which triggers a change in the screen orientation or skips a song in your playlist. These sensors are very sensitive and can measure acceleration up to 1G on a fine scale.

ALS

Function: They use of two different types of photodiodes (or light sensors) to each record the level of light currently falling on the screen. Each sensor records a different spectrum and by combining the output of these two sensors portable devices such as tablets, smart phones, and laptops can sense a side spectrum of light.

Feature: Accurately measuring the brightness of a room. This information is then used to adjust the screen brightness to make it easier on the user's eyes and also helps to extend battery life. When your smartphone is set to

auto-adjust the screen brightness, it is the ALS which tells the phone how bright/dark it is.

Barometer

Function: Measures the atmospheric pressure around the smartphone.

Feature: Pressure differences exist between small changes in altitude which is where this sensor helps. It can smartly work along with the GPS sensor to accurately provide you with your altitude information. The function wouldn't work optimally indoors since there are more factors like open/shut windows, air-conditioning which will mess with the pressure reading.

Compass

Function: The compass is simply used to figure out the magnetic north and relay the information.

Feature: Technically called a Magnetometer, the compass detects variation in magnetic field to figure out the direction. Having devices that cause electromagnetic interference can mess with the readings. This is why your compass application demands a calibration each time it is activated. Commonly used with GPS to provide map directions.

GPS

Function: Communicates with GPS satellites to determine position of the phone on the surface of the earth.

Feature: The GPS sensor works in unison with the compass and barometer to provide you with your current location. For added accuracy the GPS will also use your WiFi module and GSM antennae to locate the nearest signal antennae. GPS requires that you have as less obstruction between you and the satellites as possible.

Gyroscope

Function: Measures the angular acceleration of your smartphone while the accelerometer measures linear acceleration. Both devices are used together to get an accurate depiction of how your phone is oriented and how it came into that orientation from its previous position.

Feature: Racing games use it to allow you to use the device as a steering wheel. Other than that it is primarily used to help with the screen orientation especially when moved slowly in a manner that the accelerometer won't be able to pick up.

Humidity

Function: Measures the amount of water vapour in the air.

Feature: This is more of a luxury sensor since the ways that information regarding humidity can be used is limited. It can be used to add an extra layer of detail to your GPS sensor.

Proximity Sensor

Function: It detects how close a smartphone is to objects facing the sensor. Ideally it is an InfraRed sensor module.

Feature: It lets the smartphone know whether you have the screen close to your face when taking a call. Subsequently, the smartphone switches off the screen to save battery life. Also, with the screen shut down the phone also keeps a lot cooler, especially for calls that last a long time.

Step counter

Function: It measures the number of steps that the smartphone user takes.

Feature: Currently, the accelerometer and gyroscope are used together to figure out how many steps a person has taken over the course of a marathon or a morning jogging session. The step counter is just a specialised implementation which does the same function but consumes less power and is more accurate.

Temperature

Function: Measures the temperature around the smartphone.

Feature: Yet another superficial feature that makes your smartphone into an all-rounder information source. There could be a better implementation which would alert the authorities when the temperature goes high enough to cause a fire. 